# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The largest possible solid sphere is carved out of a cube of edge 21 cm. The volume of the material wasted is: 1

**(a)** 4410 cm³ **(b)** 6835.5 cm³ **(c)** 4851 cm³ **(d)** 9261 cm³

**2.** A solid sphere of radius 10.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). The volume of the space in the cylinder not occupied by the sphere is: 1

**(a)** 12127.5 cm³ **(b)** 4851 cm³ **(c)** 7276.5 cm³ **(d)** 2425.5 cm³

**3.** A solid cylinder of radius 6 cm and height 5 cm is melted and recast into a cone of the same base radius. The height of the cone is: 1

**(a)** 10 cm **(b)** 30 cm **(c)** 15 cm **(d)** 5 cm

**4.** A solid cone and a solid hemisphere have equal base radii 7 cm and equal volumes. The height of the cone is: 1

**(a)** 14 cm **(b)** 7 cm **(c)** 21 cm **(d)** 10.5 cm

**5.** The radii of two solid spheres are in the ratio 2 : 3. The ratio of their surface areas is: 1

**(a)** 2 : 3 **(b)** 8 : 9 **(c)** 4 : 9 **(d)** 8 : 27

**6.** The radii of two solid spheres are in the ratio 2 : 3. The ratio of their volumes is: 1

**(a)** 2 : 3 **(b)** 8 : 27 **(c)** 4 : 9 **(d)** 16 : 27

**7.** A solid cylinder of base radius 6 cm and height 10 cm is melted and recast into identical solid cones, each of base radius 3 cm and height 4 cm. The number of cones obtained is: 1

**(a)** 34 **(b)** 10 **(c)** 60 **(d)** 30

**8.** A hemispherical bowl has outer radius 9 cm and inner radius 7.5 cm. The volume of material used to make the bowl is: 1

**(a)** 643.5 cm³ **(b)** 1287 cm³ **(c)** 1930.5 cm³ **(d)** 321.75 cm³

**9.** **Assertion (A):** The total surface area of a solid hemisphere of radius r is $3\pi r^2$. 1

**Reason (R):** The total surface area of a solid hemisphere is its curved surface area $2\pi r^2$ plus the area $\pi r^2$ of its flat circular face.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**10.** **Assertion (A):** A cone and a cylinder have the same base radius and the same height. The volume of the cone is one-third that of the cylinder. 1

**Reason (R):** The volume of a cone of base radius r and height h is $\pi r^2 h$.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** A solid sphere of radius 6 cm is completely dropped into water in a cylindrical vessel of base radius 8 cm, and no water spills. Find the rise in the water level. 2

**12.** A hemispherical bowl has outer radius 7.5 cm and inner radius 4.5 cm. Find the volume of material used to make the bowl. 2

**13.** A solid metallic sphere of radius 12 cm is melted and drawn into a wire of circular cross-section of radius 2 cm. Find the length of the wire. 2

**14.** A solid sphere of radius 9 cm is completely dropped into water in a cylindrical vessel of base radius 12 cm, and no water spills. Find the rise in the water level. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A solid metallic cylinder of base radius 9 cm and height 32 cm is melted and recast into solid spheres of radius 6 cm. Find the number of spheres obtained. 3

**16.** A solid metallic cone of base radius 10 cm and height 5 cm is melted and recast into a solid sphere. Find the radius of the sphere. 3

**17.** A solid cylinder of base radius 6 cm and height 20 cm is melted and recast into identical solid cones, each of base radius 3 cm and height 8 cm. Find the number of cones obtained. 3

**OR**

A solid metallic cylinder of base radius 6 cm and height 72 cm is melted and recast into solid spheres of radius 6 cm. Find the number of spheres obtained. 3

**18.** A solid metallic cone of base radius 8 cm and height 4 cm is melted and recast into a solid sphere. Find the radius of the sphere. 3

**OR**

A solid cylinder of base radius 12 cm and height 14 cm is melted and recast into identical solid cones, each of base radius 6 cm and height 21 cm. Find the number of cones obtained. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** A medicine capsule is in the shape of a cylinder with a hemisphere stuck at each end. The radius of the cylinder and of each hemisphere is 3.5 mm and the length of the cylindrical part is 7 mm. (Take $\pi = \frac{22}{7}$.) 5

(i) Find the curved surface area of the cylindrical part.

(ii) Find the surface area of the two hemispherical ends together.

(iii) Find the total surface area of the capsule.

(iv) Find the total length of the capsule and the volume of the cylindrical part.

**20.** A toy is a cone of base radius 63 cm and height 84 cm mounted on a hemisphere of the same radius (the flat faces coincide). Find the total surface area of the toy. 5

**OR**

A solid sphere of radius 42 cm fits exactly inside a cylinder (it touches the curved surface and both bases). Find the volume of the space in the cylinder not occupied by the sphere. 5

## END OF QUESTION PAPER — Total: 40 Marks